# Chapter 8. Integer-Valued Time Series Models

*INAR(1), INGARCH(1,1), change-point tests*

## Contents

**8.1 The INAR(1) model** **2**
8.1.1 Conditional least squares . . . . . 4

**8.2 The INGARCH(1,1) model** **4**

**8.3 Change-point tests** **5**

Time series of *counts* take values in $\{0, 1, 2, \dots\}$: the number of trades of a stock per minute, monthly traffic accidents in a region, daily hospital admissions, monthly earthquakes. For such *count time series*, ARMA and GARCH models are inappropriate because $\phi y_{t-1} + \epsilon_t$ is not integer-valued. Two standard alternatives are the INAR (integer-valued autoregressive) model of McKenzie (1985) and Al-Osh and Alzaid (1987), and the INGARCH (integer-valued GARCH) model of Ferland et al. (2006), whose conditional law is Poisson with a mean depending on the past; inference for INGARCH was developed by Davis and Liu (2016), Lee and Lee (2019) and others (see Weiss, 2018). Throughout, $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ denotes the past (the book's $I_t$).

## 8.1 The INAR(1) model

**Motivation.** Multiplying an integer by $\alpha \in (0, 1)$ does not give an integer, but keeping each of $Y$ units independently with probability $\alpha$ does. This binomial thinning replaces scalar multiplication and yields an integer-valued analogue of AR(1) with the same autocorrelations.

**Definition 8.1.1 (Binomial thinning)**

For a nonnegative integer-valued random variable $Y$ and $\alpha \in (0, 1)$, the *thinning operator* $\circ$ is

$$\alpha \circ Y = \sum_{i=1}^{Y} w_i,$$

where the $w_i$ are i.i.d. Bernoulli variables with success probability $\alpha$, independent of $Y$ (the *counting sequence*); $\alpha \circ Y = 0$ when $Y = 0$.

**Proposition (Properties of thinning, [Supplement])**

Let $Y$ be nonnegative integer-valued with $EY^2 < \infty$, with probability generating function $G_Y(s) = Es^Y$, and let $\alpha, \beta \in (0, 1)$.
(1) $E(\alpha \circ Y \mid Y) = \alpha Y$ and $\operatorname{Var}(\alpha \circ Y \mid Y) = \alpha(1-\alpha)Y$; hence $E(\alpha \circ Y) = \alpha EY$ and $\operatorname{Var}(\alpha \circ Y) = \alpha(1-\alpha)EY + \alpha^2 \operatorname{Var} Y$.
(2) $G_{\alpha \circ Y}(s) = G_Y(1 - \alpha + \alpha s)$.
(3) With independent counting sequences, $\alpha \circ (\beta \circ Y) \overset{d}{=} (\alpha\beta) \circ Y$.
(4) If $Y \sim \text{Poisson}(\lambda)$ then $\alpha \circ Y \sim \text{Poisson}(\alpha\lambda)$.

**Proof.**

(1) Given $Y = y$, $\alpha \circ Y$ is Binomial$(y, \alpha)$, with mean $\alpha y$ and variance $\alpha(1-\alpha)y$. The unconditional formulas follow from $E(\cdot) = E(E(\cdot \mid Y))$ and $\operatorname{Var}(\cdot) = E \operatorname{Var}(\cdot \mid Y) + \operatorname{Var} E(\cdot \mid Y)$.

(2) $E(s^{\alpha \circ Y} \mid Y) = (Es^{w_1})^Y = (1 - \alpha + \alpha s)^Y$; take expectations.

(3) By (2) twice, $G_{\alpha \circ (\beta \circ Y)}(s) = G_{\beta \circ Y}(1-\alpha+\alpha s) = G_Y\big(1-\beta+\beta(1-\alpha+\alpha s)\big) = G_Y(1-\alpha\beta+\alpha\beta s)$, the generating function of $(\alpha\beta) \circ Y$.

(4) $G_Y(s) = e^{\lambda(s-1)}$, so $G_{\alpha \circ Y}(s) = e^{\lambda(1-\alpha+\alpha s-1)} = e^{\alpha\lambda(s-1)}$. ∎

**Definition 8.1.2 (INAR(1) model)**

A series $y_t$, $t = 1, 2, \dots$, with values in $\{0, 1, 2, \dots\}$ follows the *INAR(1) model* if

$$y_t = \alpha \circ y_{t-1} + \epsilon_t, \tag{8.1}$$

where $\alpha \in (0,1)$, the $\epsilon_t$ are i.i.d. nonnegative integer-valued random variables, and $\epsilon_t$ and the counting sequence of $\alpha \circ y_{t-1}$ are independent of each other and of $\mathcal{F}_{t-1}$. The term $\alpha \circ y_{t-1}$ is closely related to a branching process (each unit present at $t-1$ survives to $t$ with probability $\alpha$), and $\epsilon_t$ can be viewed as the number of immigrants.

Write $\mu = E\epsilon_t$ and $\sigma_\epsilon^2 = \operatorname{Var}\epsilon_t$. Usually $\epsilon_t$ is taken Poisson or geometric.

**Proposition (Stationarity, moments and ACF of INAR(1), [Supplement])**

Let $0 < \alpha < 1$ and $\sigma_\epsilon^2 < \infty$.
(1) (8.1) has a strictly stationary solution; every stationary solution has the marginal law of $\sum_{j=0}^{\infty} \alpha^j \circ \epsilon_{t-j}$ (independent thinnings), which is finite a.s.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = \alpha y_{t-1} + \mu$ and $\operatorname{Var}(y_t \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha)y_{t-1} + \sigma_\epsilon^2$.
(3) $Ey_t = \dfrac{\mu}{1-\alpha}$ and $\operatorname{Var} y_t = \dfrac{\sigma_\epsilon^2 + \alpha\mu}{1-\alpha^2}$.
(4) $\gamma(h) = \operatorname{Cov}(y_t, y_{t+h}) = \alpha^{|h|}\gamma(0)$, so the ACF is $\rho(h) = \alpha^{|h|}$, as for an AR(1) process.

**Proof.**

*Step 1: construction.* Take i.i.d. $\epsilon_s$ ($s \in \mathbb{Z}$) and, independently, i.i.d. Bernoulli($\alpha$) indicators $B_{s,i,r}$ for $s < r$ and $i \geq 1$. Immigrant $i$ of time $s$ ($1 \leq i \leq \epsilon_s$) is *alive at time* $t \geq s$ if $B_{s,i,r} = 1$ for all $s < r \leq t$, and let $y_t$ be the number of individuals alive at time $t$. Since $Ey_t = \sum_{j\geq 0} \alpha^j \mu < \infty$, $y_t$ is finite a.s. Each individual alive at $t-1$ survives to $t$ independently with probability $\alpha$, via indicators $B_{\cdot,\cdot,t}$ independent of $\mathcal{F}_{t-1}$ and of $\epsilon_t$; so the survivors form $\alpha \circ y_{t-1}$ and $y_t = \alpha \circ y_{t-1} + \epsilon_t$. The construction is shift-invariant, so $\{y_t\}$ is strictly stationary. The contributions of the cohorts $s = t-j$ are independent, and each is distributed as $\alpha^j \circ \epsilon_{t-j}$ (survival over $j$ steps has probability $\alpha^j$); hence $y_t \overset{d}{=} \sum_{j\geq 0} \alpha^j \circ \epsilon_{t-j}$.

*Step 2: uniqueness of the law.* For any solution, iterating (8.1) and using Proposition (3) gives $y_t \overset{d}{=} \alpha^t \circ y_0 + \sum_{j=0}^{t-1} \alpha^j \circ \epsilon_{t-j}$ with all thinnings independent. As $P(\alpha^t \circ y_0 \geq 1) = E\left[1-(1-\alpha^t)^{y_0}\right] \to 0$ by dominated convergence, the law of $y_t$ converges to that of $\sum_{j\geq 0} \alpha^j \circ \epsilon_{t-j}$. If the solution is stationary its law does not change with $t$, so it equals this limit.

*Step 3: conditional moments.* Given $\mathcal{F}_{t-1}$, $\alpha \circ y_{t-1}$ is Binomial($y_{t-1}, \alpha$) and independent of $\epsilon_t$, so Proposition (1) gives (2); in particular

$$E(y_t \mid y_{t-1}) = \alpha y_{t-1} + \mu. \tag{8.2}$$

*Step 4: moments.* The representation of Step 1 has finite variance, since $\operatorname{Var}(\alpha^j \circ \epsilon) = \alpha^j(1-\alpha^j)\mu + \alpha^{2j}\sigma_\epsilon^2$ is summable. Taking expectations in (8.2), $m = Ey_t$ solves $m = \alpha m + \mu$. By the variance decomposition and (2),

$$v = \operatorname{Var} y_t = E(\alpha(1-\alpha)y_{t-1} + \sigma_\epsilon^2) + \operatorname{Var}(\alpha y_{t-1} + \mu) = \alpha(1-\alpha)m + \sigma_\epsilon^2 + \alpha^2 v,$$

and $(1-\alpha)m = \mu$ gives $v(1-\alpha^2) = \alpha\mu + \sigma_\epsilon^2$.

*Step 5: ACF.* For $h \geq 1$, $y_t$ is $\mathcal{F}_{t+h-1}$-measurable, so $\gamma(h) = \operatorname{Cov}(E(y_{t+h} \mid \mathcal{F}_{t+h-1}), y_t) = \operatorname{Cov}(\alpha y_{t+h-1} + \mu, y_t) = \alpha\gamma(h-1)$, whence $\gamma(h) = \alpha^h\gamma(0)$; and $\gamma(-h) = \gamma(h)$. ∎

**Example 8.1.3 (Poisson innovations)**

Let $\epsilon_t \sim \text{Poisson}(\mu)$. By Proposition (4), $\alpha^j \circ \epsilon_{t-j} \sim \text{Poisson}(\alpha^j\mu)$, and a sum of independent Poisson variables is Poisson, so the stationary law is

$$y_t \sim \text{Poisson}\Big(\frac{\mu}{1-\alpha}\Big),$$

with mean equal to variance, consistent with Proposition (3) for $\sigma_\epsilon^2 = \mu$. Larger $\alpha$ gives a higher level and longer runs of similar counts, the autocorrelation at lag $h$ being $\alpha^h$.

### 8.1.1 Conditional least squares

By (8.2), $(\alpha, \mu)$ can be estimated by least squares: given $y_0, y_1, \ldots, y_n$, $(\hat\alpha, \hat\mu)$ minimizes $\sum_{t=1}^n (y_t - \alpha y_{t-1} - \mu)^2$. Solving the normal equations,

$$\begin{pmatrix} \hat\alpha \\ \hat\mu \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^n y_{t-1}^2 & \sum_{t=1}^n y_{t-1} \\ \sum_{t=1}^n y_{t-1} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{t=1}^n y_{t-1} y_t \\ \sum_{t=1}^n y_t \end{pmatrix}. \tag{8.3}$$

**Proposition (Asymptotics of the least squares estimator, [Supplement])**

Let $\{y_t\}$ be the stationary INAR(1) series with $\operatorname{Var} y_t > 0$ and $Ey_t^3 < \infty$. Put $\theta = (\alpha, \mu)^T$, $\mathbf{x}_t = (y_{t-1}, 1)^T$, $u_t = y_t - \alpha y_{t-1} - \mu$, $A = E(\mathbf{x}_t \mathbf{x}_t^T)$ and $V = E(u_t^2 \mathbf{x}_t \mathbf{x}_t^T)$. Then $\hat\theta \to \theta$ a.s. and

$$\sqrt{n}(\hat\theta - \theta) \xrightarrow{d} N(0, A^{-1} V A^{-1}).$$

**Proof.**

*Step 1.* $\{y_t\}$ is a fixed function of an i.i.d. sequence (Step 1 above), hence stationary and ergodic. $A$ is positive definite because $\det A = \operatorname{Var} y_{t-1} > 0$. Since $y_t = \theta^T \mathbf{x}_t + u_t$, (8.3) gives

$$\hat\theta - \theta = \Big( \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t^T \Big)^{-1} \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t u_t.$$

*Step 2.* By the ergodic theorem $\frac{1}{n}\sum \mathbf{x}_t \mathbf{x}_t^T \to A$ and $\frac{1}{n}\sum \mathbf{x}_t u_t \to E(\mathbf{x}_t u_t) = E\big(\mathbf{x}_t E(u_t \mid \mathcal{F}_{t-1})\big) = 0$ a.s., so $\hat\theta \to \theta$ a.s.

*Step 3.* $\{\mathbf{x}_t u_t\}$ is a stationary ergodic martingale difference with $E\|\mathbf{x}_t u_t\|^2 < \infty$ (because $E(u_t^2 \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha) y_{t-1} + \sigma_\epsilon^2$ and $Ey^3 < \infty$). The martingale CLT gives $n^{-1/2} \sum \mathbf{x}_t u_t \xrightarrow{d} N(0, V)$, and Slutsky's lemma completes the proof. ∎

## 8.2 The INGARCH(1,1) model

**Motivation.** Whereas INAR models the count itself, INGARCH models its conditional mean: the count is Poisson given the past, with an intensity following a GARCH-type recursion. Because the Poisson variance equals the mean, this is simultaneously a model for the conditional variance.

Suppose the conditional mean of $y_t$ given the past follows the recursion

$$\mu_t = \omega + \alpha y_{t-1} + \beta \mu_{t-1}, \qquad t = 1, \ldots, n, \tag{8.4}$$

where $\omega > 0$, $\alpha, \beta > 0$ with $\alpha + \beta < 1$, and the initial values $y_0$ (a nonnegative integer) and $\mu_0$ (a positive real) are arbitrary, and that $y_t$ is generated from a Poisson distribution with parameter $\mu_t$:

$$P(y_t = k \mid \mathcal{F}_{t-1}) = \frac{e^{-\mu_t} \mu_t^k}{k!}, \qquad k = 0, 1, \ldots \tag{8.5}$$

**Definition 8.2.1 (INGARCH(1,1) model)**

Combining (8.4) and (8.5), the *INGARCH(1,1) model* is

$$y_t \mid \mathcal{F}_{t-1} \sim \text{Poisson}(\mu_t), \qquad \mu_t = \omega + \alpha y_{t-1} + \beta \mu_{t-1}. \tag{8.6}$$

By the Poisson property, the conditional variance of $y_t$ also equals $\mu_t$.

Writing (8.6) with the backshift operator, $(1-\beta B)\mu_t = \omega + \alpha y_{t-1}$, so

$$\mu_t = \frac{\omega}{1-\beta} + \alpha \sum_{j=1}^{\infty} \beta^{j-1} y_{t-j}, \tag{8.7}$$

and $\mu_t$ is a function of $(\omega, \alpha, \beta)$ and the past counts. The analogy with GARCH(1,1) is exact: $\sigma_t^2 \leftrightarrow \mu_t$, $y_{t-1}^2 \leftrightarrow y_{t-1}$.

**Proposition (Moments of INGARCH(1,1), [Supplement])**

Let $\{y_t\}$ be a second-order stationary solution of (8.6) (Ferland et al., 2006, show that one exists when $\alpha + \beta < 1$). Let $m = Ey_t$. Then
(1) $m = E\mu_t = \dfrac{\omega}{1-\alpha-\beta}$;
(2) $\operatorname{Var} \mu_t = \dfrac{\alpha^2 m}{1-(\alpha+\beta)^2}$ and $\operatorname{Var} y_t = m \dfrac{1-(\alpha+\beta)^2+\alpha^2}{1-(\alpha+\beta)^2} \geq m$, so the counts are overdispersed;
(3) for $h \geq 1$, $\operatorname{Cov}(y_t, y_{t-h}) = (\alpha+\beta)^{h-1}(\alpha \operatorname{Var} y_t + \beta \operatorname{Var} \mu_t)$: the ACF decays geometrically at rate $\alpha+\beta$, like that of an ARMA(1,1).

**Proof.** Let $\nu_t = y_t - \mu_t$. Since $\mu_t$ is $\mathcal{F}_{t-1}$-measurable, $E(\nu_t \mid \mathcal{F}_{t-1}) = 0$ and $E(\nu_t^2 \mid \mathcal{F}_{t-1}) = \mu_t$; hence $\nu_t$ is uncorrelated with every square-integrable $\mathcal{F}_{t-1}$-measurable variable, and $E\nu_t^2 = m$.

*Step 1.* Taking expectations in (8.4), $m = \omega + (\alpha+\beta)m$ (note $E\mu_t = Ey_t$).

*Step 2.* Substituting $y_{t-1} = \mu_{t-1} + \nu_{t-1}$, $\mu_t = \omega + (\alpha+\beta)\mu_{t-1} + \alpha\nu_{t-1}$, where $\mu_{t-1}$ and $\nu_{t-1}$ are uncorrelated. Hence $\operatorname{Var} \mu_t = (\alpha+\beta)^2 \operatorname{Var} \mu_{t-1} + \alpha^2 m$, which by stationarity gives the formula for $\operatorname{Var} \mu_t$. Then $\operatorname{Var} y_t = \operatorname{Var} \mu_t + E\nu_t^2 = \operatorname{Var} \mu_t + m$.

*Step 3.* For $h \geq 1$, $y_{t-h}$ is $\mathcal{F}_{t-1}$-measurable, so $\operatorname{Cov}(y_t, y_{t-h}) = \operatorname{Cov}(\mu_t, y_{t-h})$. For $h = 1$: $\operatorname{Cov}(\omega + \alpha y_{t-1} + \beta\mu_{t-1}, y_{t-1}) = \alpha \operatorname{Var} y_{t-1} + \beta \operatorname{Cov}(\mu_{t-1}, \mu_{t-1} + \nu_{t-1}) = \alpha \operatorname{Var} y + \beta \operatorname{Var} \mu$. For $h \geq 2$: $\operatorname{Cov}(\mu_t, y_{t-h}) = \operatorname{Cov}((\alpha+\beta)\mu_{t-1} + \alpha\nu_{t-1}, y_{t-h}) = (\alpha+\beta)\operatorname{Cov}(\mu_{t-1}, y_{t-h}) = (\alpha+\beta)\operatorname{Cov}(y_{t-1}, y_{t-h})$, using that $\nu_{t-1}$ is uncorrelated with $y_{t-h}$. Induction on $h$ finishes the proof. ■

Given $y_1, \ldots, y_n$, the conditional maximum likelihood estimator of $(\omega, \alpha, \beta)$ maximizes the conditional likelihood or, equivalently, the log-likelihood (up to a constant)

$$L(\omega, \alpha, \beta) = \prod_{t=1}^{n} \frac{e^{-\mu_t} \mu_t^{y_t}}{y_t!}, \qquad l(\omega, \alpha, \beta) = \sum_{t=1}^{n} (y_t \log \mu_t - \mu_t),$$

with $\mu_t$ computed recursively from (8.4). The estimator $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ is consistent and asymptotically normal (Lee and Lee, 2019). Besides the Poisson, the negative binomial distribution and, more generally, one-parameter exponential families are used as conditional distributions.

## 8.3 Change-point tests

**Motivation.** As for GARCH, the parameters of count models may shift; a CUSUM test on the residuals, which form a martingale difference sequence under the null, detects the change.

The test applies the CUSUM idea of Section 6.5 to residuals. For INAR(1),

$$\hat{\epsilon}_t = y_t - \hat{\alpha} y_{t-1} - \hat{\mu}, \tag{8.8}$$

and for INGARCH(1,1),

$$\hat{\epsilon}_t = y_t - \hat{\mu}_t, \qquad \hat{\mu}_t = \hat{\omega} + \hat{\alpha} y_{t-1} + \hat{\beta}\hat{\mu}_{t-1}, \tag{8.9}$$

with arbitrary initial values $y_0, \hat{\mu}_0$. The residual-based CUSUM statistic is

$$\hat{T}_n = \max_{1 \le k \le n} \frac{1}{\sqrt{n}\,\hat{\tau}} \Big| \sum_{t=1}^{k} \hat{\epsilon}_t - \frac{k}{n} \sum_{t=1}^{n} \hat{\epsilon}_t \Big|, \qquad \hat{\tau}^2 = \frac{1}{n}\sum_{t=1}^{n} \hat{\epsilon}_t^2 - \Big(\frac{1}{n}\sum_{t=1}^{n} \hat{\epsilon}_t\Big)^2. \tag{8.10}$$

At the 5% level a change is declared if $\hat{T}_n > 1.358$, and the maximizing $k$ is taken as the change point.

**Proposition (Null limit of the residual CUSUM test, [Supplement])**

Let $\{y_t\}$ be a stationary INAR(1) series satisfying the assumptions of Proposition , with no parameter change, and let $\hat{\epsilon}_t$ be the residuals (8.8) from the least squares estimator. Then $\hat{T}_n \xrightarrow{d} \sup_{0 \le u \le 1} |B^0(u)|$, the Brownian-bridge limit of Proposition .

**Proof.**

*Step 1: true innovations.* The true residuals $u_t = y_t - \alpha y_{t-1} - \mu = (\alpha \circ y_{t-1} - \alpha y_{t-1}) + (\epsilon_t - \mu)$ form a stationary ergodic martingale difference sequence with variance $\tau^2 = E u_t^2 = \alpha(1-\alpha)Ey_t + \sigma_\epsilon^2 > 0$. By the functional CLT for such martingale differences (Billingsley), $W_n(s) = \frac{1}{\tau\sqrt{n}} \sum_{t \le \lfloor ns \rfloor} u_t \xrightarrow{d} W$. Exactly as in the proof of Proposition (Steps 2–3), $\max_k \frac{1}{\tau\sqrt{n}} \big| \sum_{t \le k} u_t - \frac{k}{n} \sum_{t \le n} u_t \big| \xrightarrow{d} \sup_s |B^0(s)|$.

*Step 2: estimation effect.* $\hat{\epsilon}_t - u_t = (\alpha - \hat{\alpha}) y_{t-1} + (\mu - \hat{\mu})$. Constants cancel in $\sum_{t \le k} c - \frac{k}{n} \sum_{t \le n} c = 0$, so

$$\frac{1}{\sqrt{n}} \Big| \sum_{t \le k} (\hat{\epsilon}_t - u_t) - \frac{k}{n} \sum_{t \le n} (\hat{\epsilon}_t - u_t) \Big| = \sqrt{n} |\hat{\alpha} - \alpha| \cdot \frac{1}{n} \Big| \sum_{t \le k} (y_{t-1} - m) - \frac{k}{n} \sum_{t \le n} (y_{t-1} - m) \Big|,$$

with $m = Ey_t$. Here $\sqrt{n}(\hat{\alpha} - \alpha) = O_p(1)$ by Proposition , and $\max_{k \le n} \frac{1}{n} |\sum_{t \le k} (y_{t-1} - m)| \to 0$ a.s. because $\frac{1}{k} \sum_{t \le k} (y_{t-1} - m) \to 0$ a.s. (ergodic theorem). So the maximum over $k$ of this difference is $o_p(1)$.

*Step 3: variance estimate.* Similarly $\hat{\epsilon}_t = u_t + O(|\hat{\alpha} - \alpha| y_{t-1} + |\hat{\mu} - \mu|)$ and the ergodic theorem give $\hat{\tau}^2 \to \tau^2$ in probability. Slutsky's lemma combines Steps 1–3. ∎

For INGARCH(1,1) the residuals $y_t - \mu_t$ are again martingale differences, and the same limit holds for the residuals (8.9) under the regularity conditions of Lee and Lee (2019); the critical value 1.358 is thus used in both models (cf. Section 6.5).